Charle Maro

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Professional Summary

Software Engineer with professional experience developing, debugging, testing, and maintaining Java backend software for an AI-powered automation platform. Strong foundation in software engineering, object-oriented programming, data structures and algorithms, debugging, code quality, maintainability, and performance optimization. Experienced in analyzing application behavior, troubleshooting production issues, optimizing database queries and APIs, and developing reliable RESTful services. Hands-on experience with Java, Spring Boot, MongoDB, MySQL, Selenium, Windows desktop automation, and performance-focused debugging.

Technical Skills

- **Programming Languages:** Java, Python, SQL, JavaScript
- **Software Engineering:** Object-Oriented Programming, Data Structures and Algorithms, Debugging, Problem Solving, Code Quality, Maintainability, Testing
- **Backend Development:** Spring Boot, Spring MVC, RESTful APIs, JSON, HTTP/HTTPS, Microservices
- **Databases:** MongoDB, MySQL, PostgreSQL; Spring Data MongoDB, JPA, Hibernate
- **Automation & Systems:** Selenium 4, Appium, WinAppDriver, Windows Desktop Automation, WebDriver, System-Level Debugging
- **Performance:** Application Performance Analysis, API Optimization, Database Query Optimization, Production Debugging
- **Tools:** Git, GitHub, Postman, IntelliJ IDEA, CI/CD Pipelines

Experience

FireFlink June 2025 – Present
Associate Software Developer *Bengaluru, India*

- Develop, test, debug, and maintain enterprise software using **Java and Spring Boot** for an AI-powered test automation platform, following object-oriented design and established software engineering practices.
- Investigate and debug application and automation failures across **Windows desktop, web, and mobile environments**, analyzing execution behavior and logs to identify root causes and implement reliable fixes.
- Diagnose system-level automation issues involving **Windows applications, WinAppDriver, Selenium, WebDriver, and application processes**, improving compatibility and execution reliability.
- Optimize backend services, database queries, and application workflows using profiling, debugging, log analysis, and performance-focused investigation to improve application stability.
- Design and implement maintainable **RESTful APIs** using Spring MVC and Spring Boot for internal automation, workflow, and configuration services.
- Build and maintain data persistence components using **Spring Data MongoDB**, including query optimization, schema design, data validation, and backend performance improvements.
- Implement **Spring Security** for authentication, authorization, and role-based access control while maintaining secure and reliable backend behavior.
- Investigate production defects, perform root-cause analysis, validate fixes through testing, and incorporate feedback from code reviews to maintain high-quality software.
- Developed a unified defect management integration with **ClickUp APIs**, enabling defect creation, synchronization, and status tracking across ClickUp, Azure DevOps, and Jira through a single platform.
- Collaborate with QA, frontend, and platform teams to analyze issues, communicate technical findings, and deliver reliable software improvements in an Agile development environment.

Projects

Java-Based Blockchain System June 2023

- Developed a Java-based blockchain system to implement core concepts of distributed data structures, cryptographic hashing, transaction processing, and data integrity.
- Applied object-oriented programming, data structures, algorithmic problem solving, and software engineering principles to design and implement core blockchain components.

E-Commerce Backend Application 2025

- Built a backend application using **Java, Spring Boot, and Spring MVC** with RESTful APIs for user management, product catalog, orders, and payment workflows.
- Implemented **Spring Security** for authentication and authorization and used JPA/Hibernate for relational data modeling and persistence.
- Tested backend functionality and applied maintainable service-layer design, data validation, exception handling, and reliable API development practices.

Education

Panimalar Institute of Technology, Chennai May 2025
B.E. in Computer Science and Engineering *GPA: 8.35 / 10*